1.2) Along the long axis, the sensor outputting above 2.50 V near the pole face and decreases toward 2.50 V when distance increase. At large distances (d >> magnet length), the field follows an approximate 1/d^3 decay following with magnetic dipole behaviour.

Along the short axis, the output remains very close to 2.50 V at all distances, which shows a near-zero detected field component. This is because the magnetic field lines start from the N pole and curve around to the S pole; at the side of the magnet, the field lines are mostly parallel to the long axis. The sensor detects only the component perpendicular to its sensing surface, which is minimal along the short axis direction.

2.1) When the current-carrying coil is at rest, it sits at an equilibrium position where the net magnetic torque is zero which means the coil's magnetic moment is parallel to the external field B, so the torque is zero. Tapping the coil provides a small initial momentum to the coil so that the coil moves away from the zero-torque position. Once displaced, the torque becomes nonzero and then the coil will start continuous rotation.

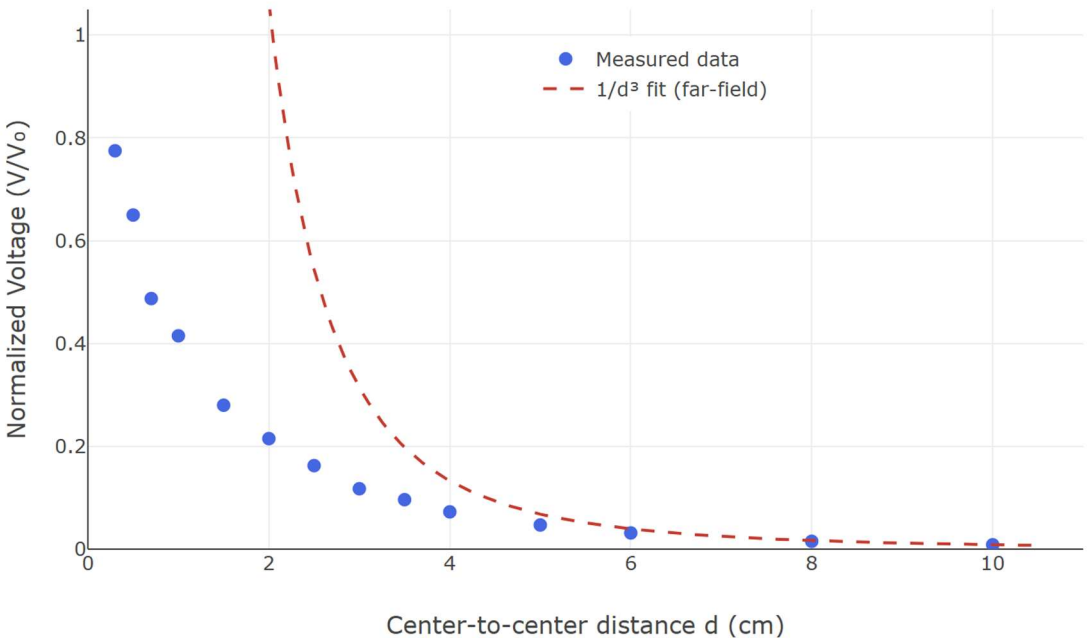
2.3) The coil can keep rotating in one direction because of the split-ring commutator. It reverses the current direction through the coil every half turn as the coil passes through its equilibrium position. This ensures the torque always acts in the same rotational direction, so instead of oscillating back and forth, the coil spins continuously.

2.5) Yes, it agrees with expectation. The magnet has its north pole pointing upward, so B is directed vertically upward through the coil. The DC supply's positive terminal is connected to the left and the negative to the right, establishing a specific current direction around the coil loop. Applying the Lorentz force F = IL x B(right-hand rule) to each current-carrying segment of the coil, the forces on opposite sides form a couple whose torque direction is consistent with anticlockwise rotation when viewed from the right, matching the observation.

3.3) No, they are not equal.

The magnetic coupling between the two coils is imperfect as the coupling coefficient is less than 1 because not all magnetic flux generated by the first coil passes through the second coil; some flux leaks into the surrounding space. The induced voltage on CH2 is therefore lower than the driving voltage on CH1.

3.4) The normalized CH2 voltage decreases rapidly as the distance between the two coils increases. This is because the mutual inductance M between the coils decreases with separation, and the induced EMF is directly proportional to M. At large separations where d >> a, each coil behaves as a magnetic dipole whose on-axis field decays as 1/d^3, so the induced voltage follows the same approximate dependence. At shorter distances, the measured voltage falls below the 1/d^3 prediction because the finite coil geometry limits the maximum coupling and the near-field flux distribution deviates from the simple dipole model.



3.7) The measured voltage ratio is 2, which matches the turns ratio 54/27 = 2. Since both receiving coils are placed at the same position, the flux per turn is identical, so the induced EMF scales linearly with N_2. Doubling the turns exactly doubles the voltage.